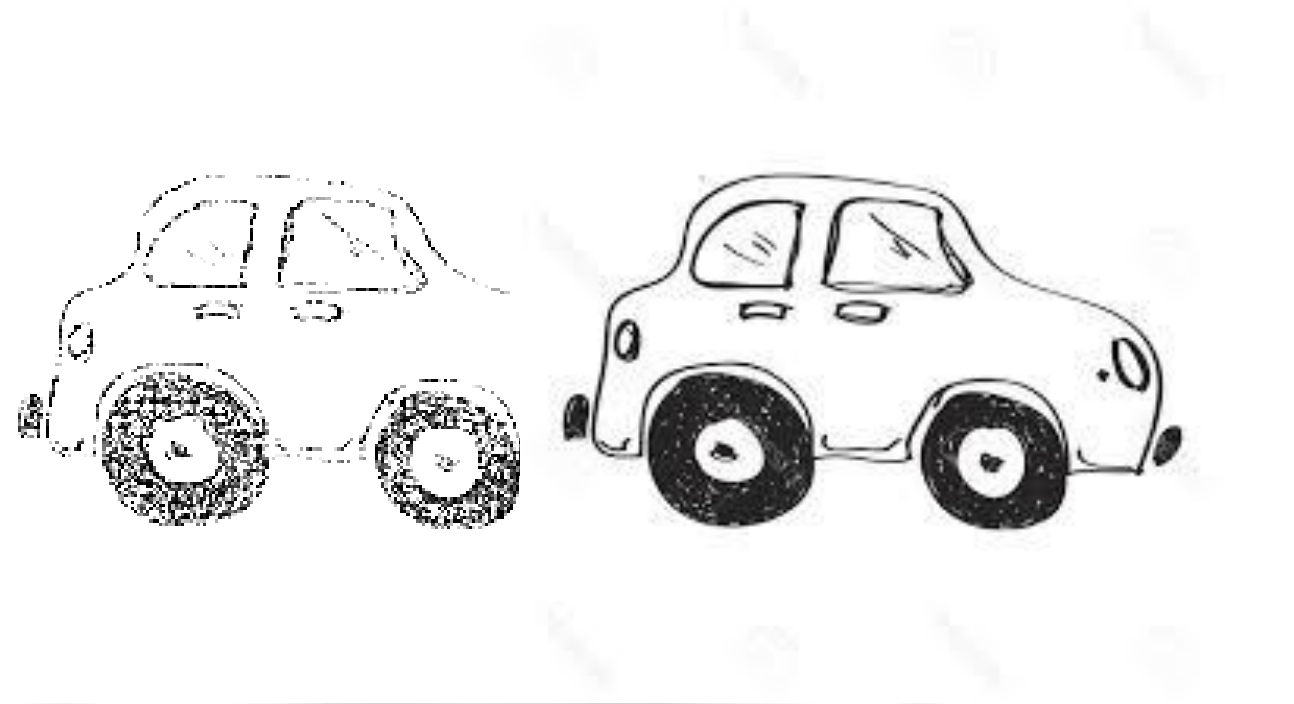
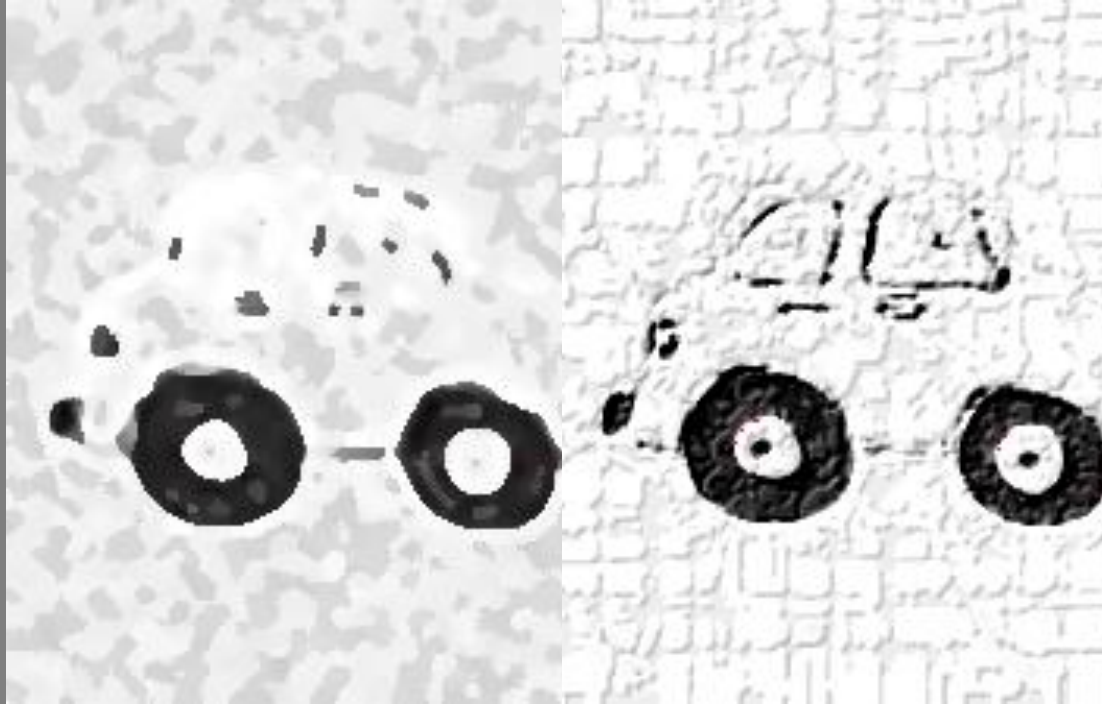


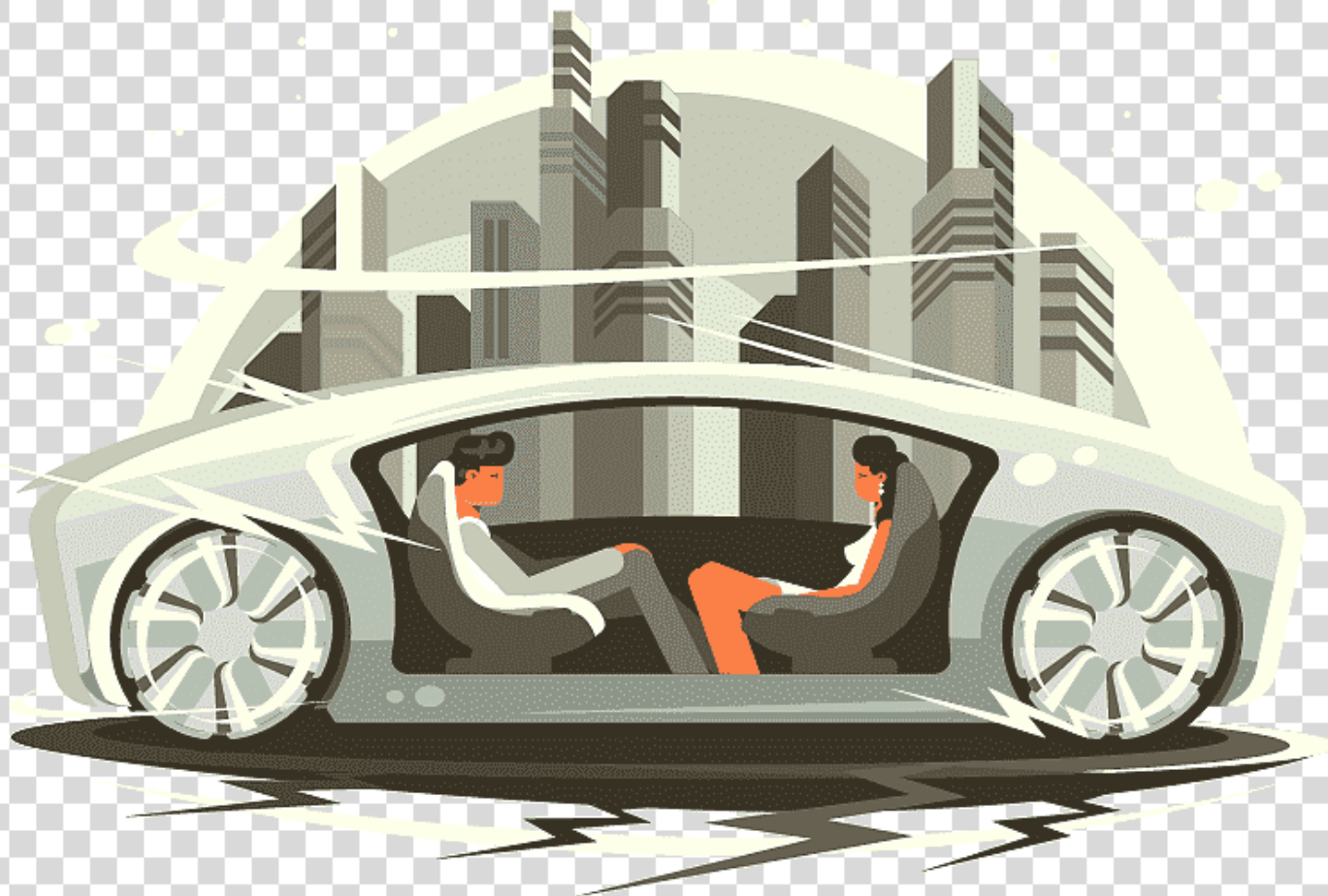


ROBOTI MOBILI

Cuprins

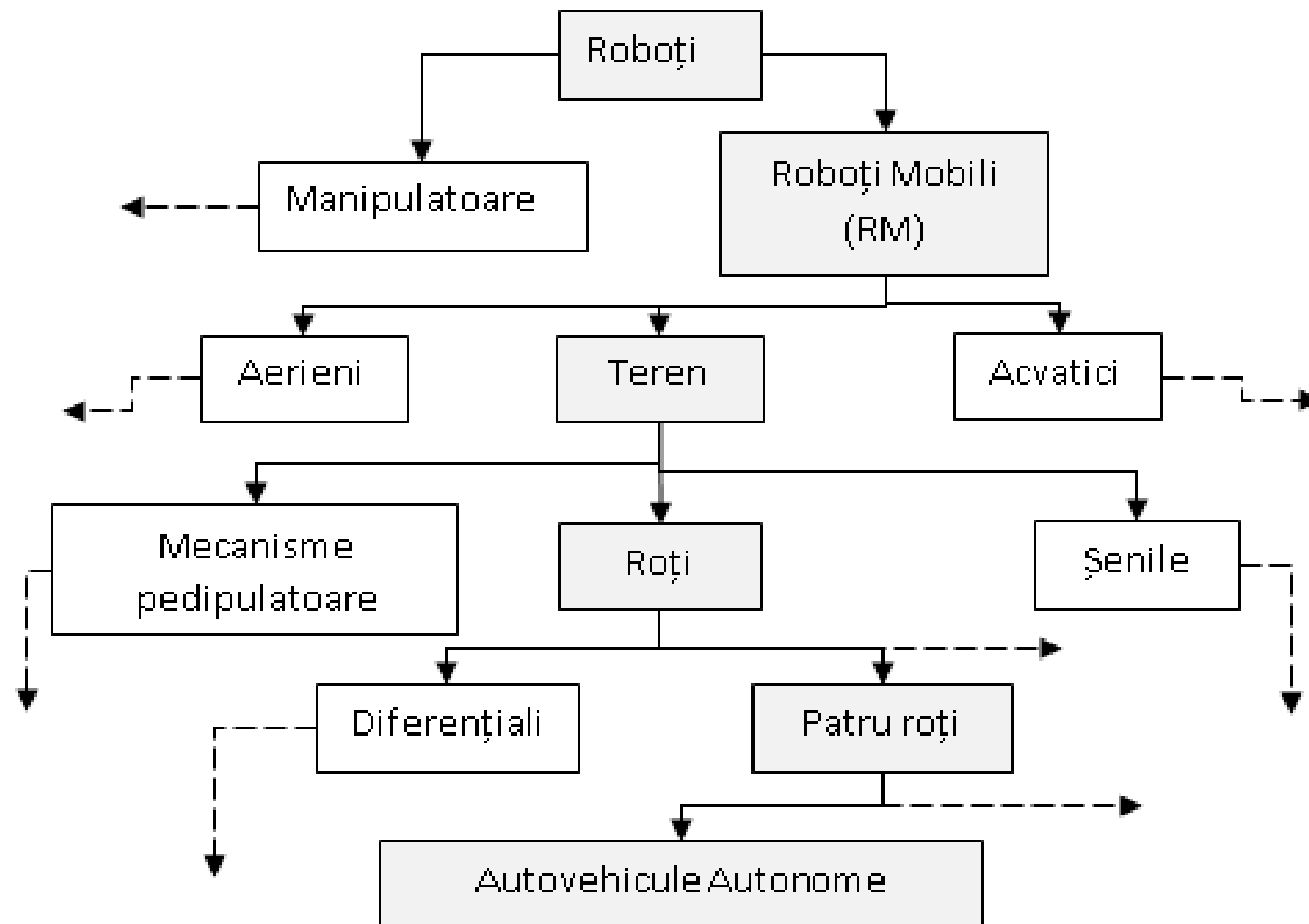
- ***Obiectivele modulului;***
- ***Structura modulului;***
- ***Bibliografie;***
- ***Programe utilizate;***
- ***Despre cauzalitate.***

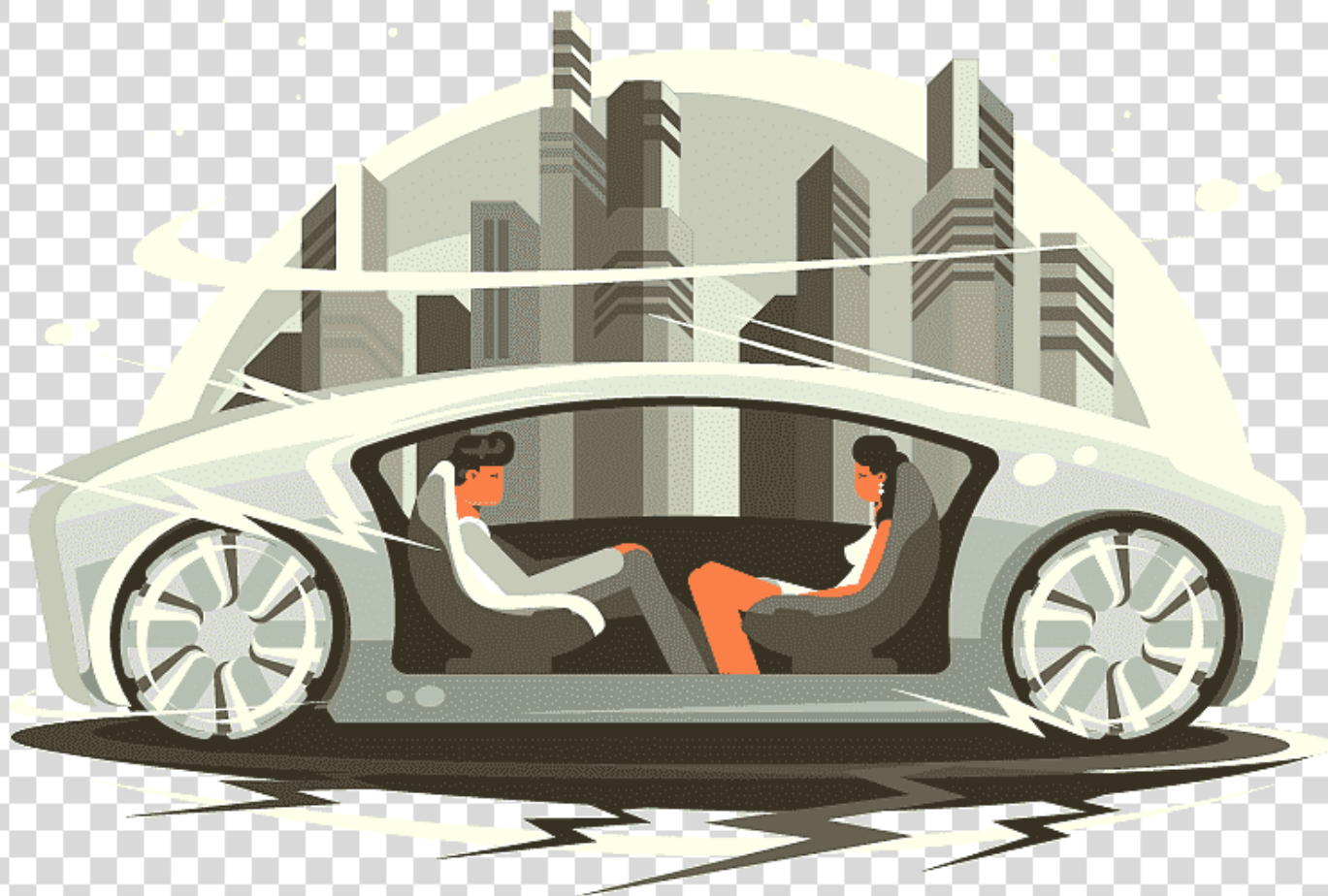




Obiectivele modului

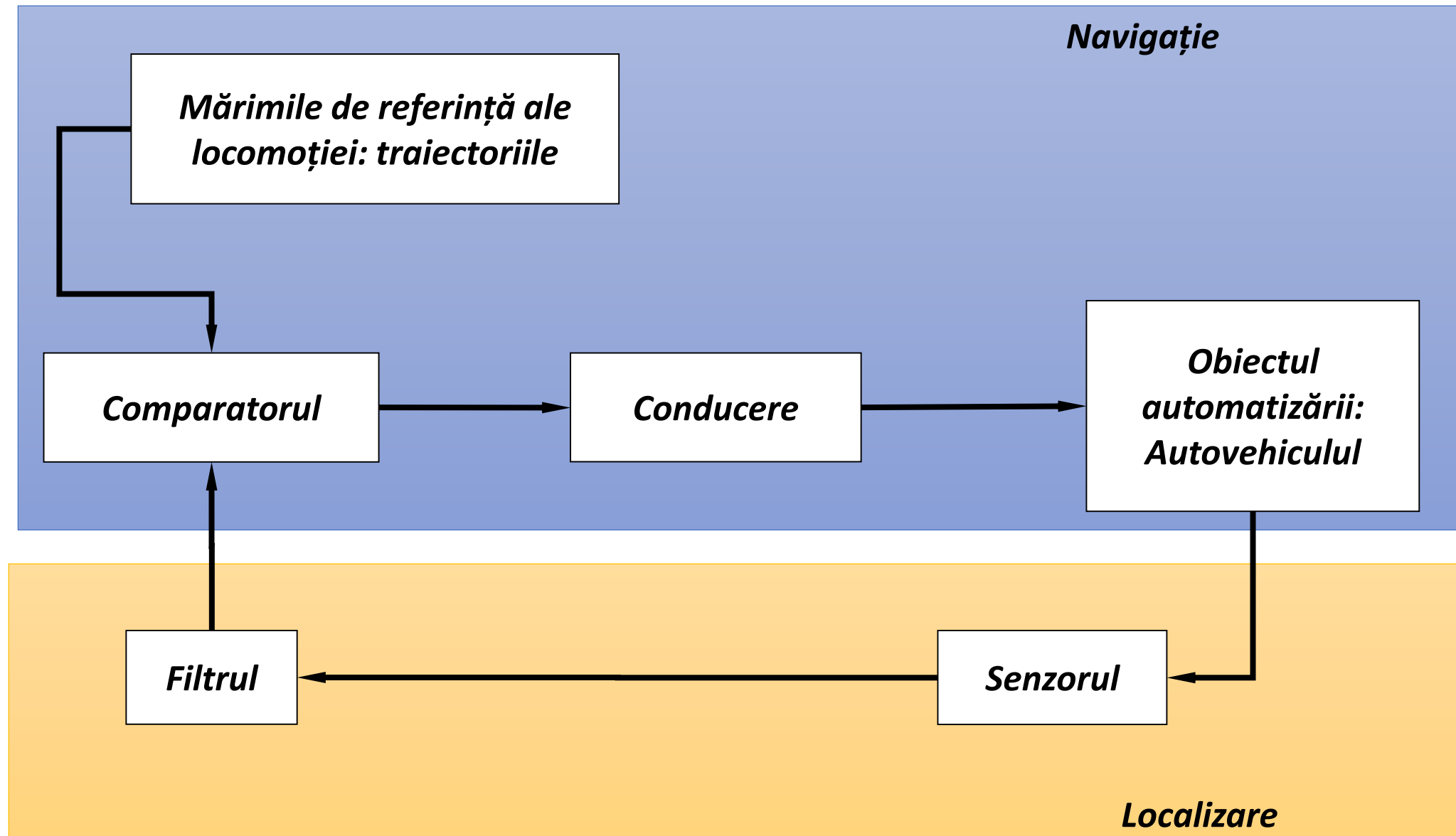
Obiectivul fundamental al acestui modul este **prezentarea procedeeelor de modelare (descriere matematică) a roboților mobili cu roți - a autovehiculelor autonome**. Se oferă cunoștințe fundamentale referitoare la RM transpuse apoi în algoritmi și programe de calcul.

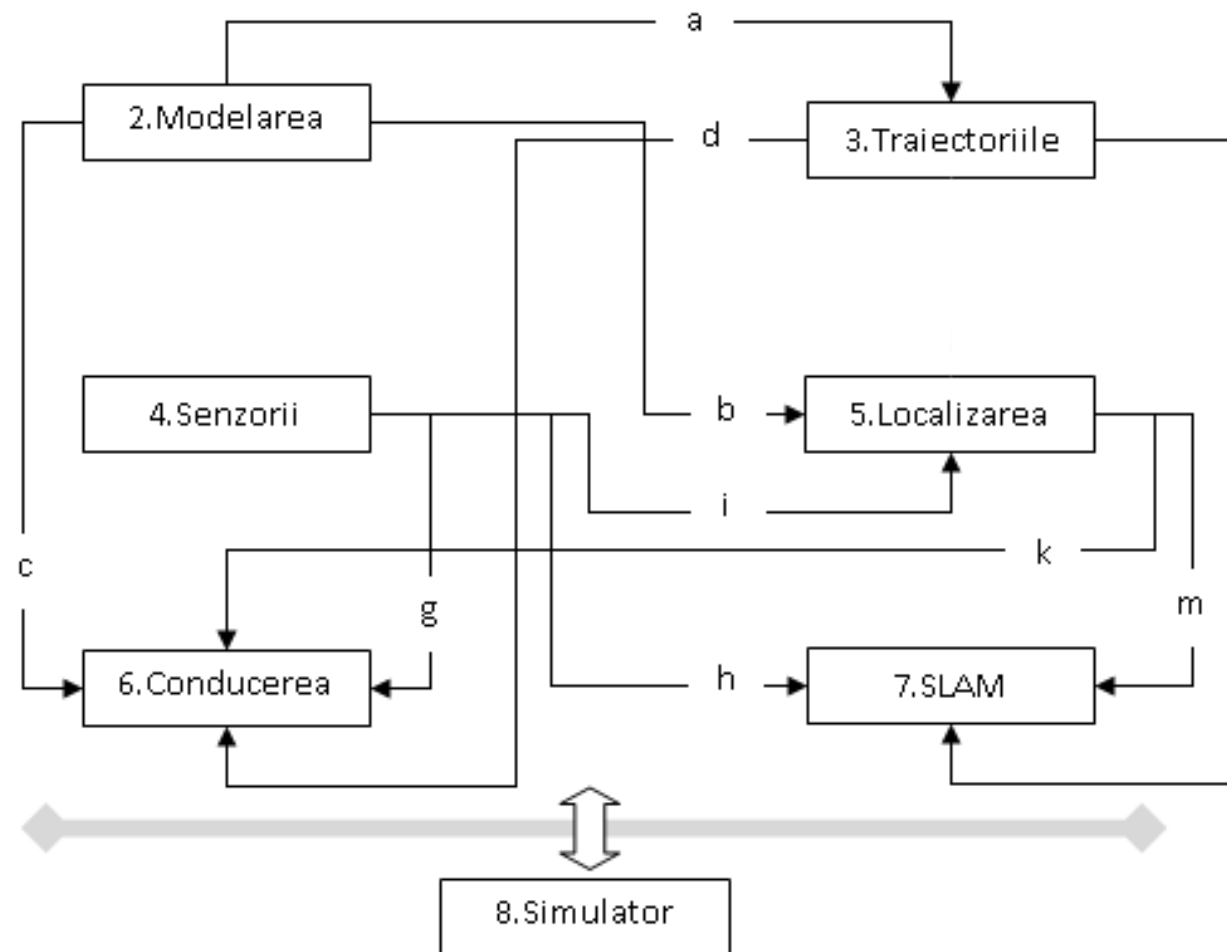




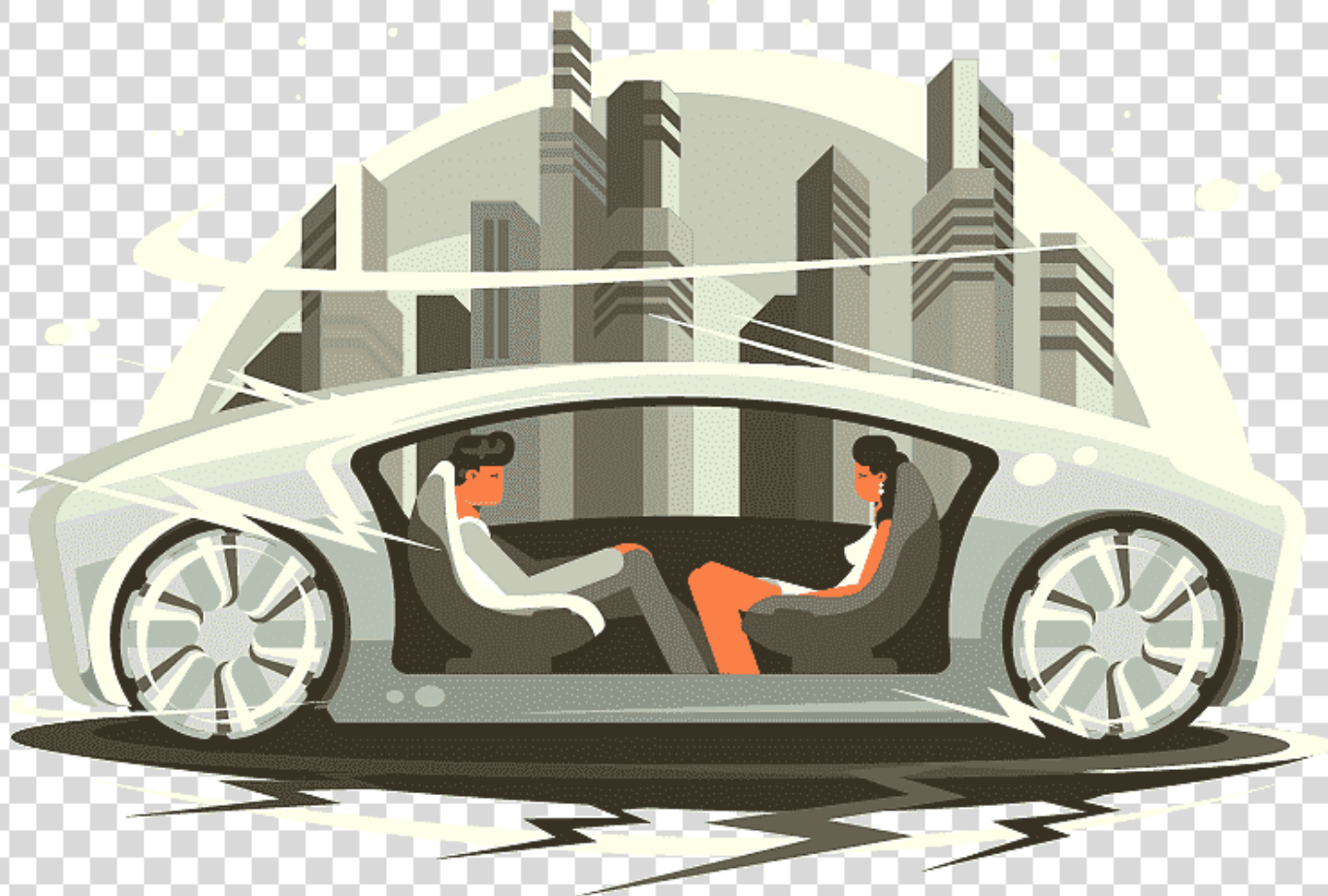
8 Capítole

Structura modulului





Capitolul	Modelare			Traietorii		Senzori		Localizare			Conducere		SLAM		
Săptămâna	9			10		10		11	12		13	14			



Bibliografie



Modelarea robotului mobil de tip autovehicul autonom

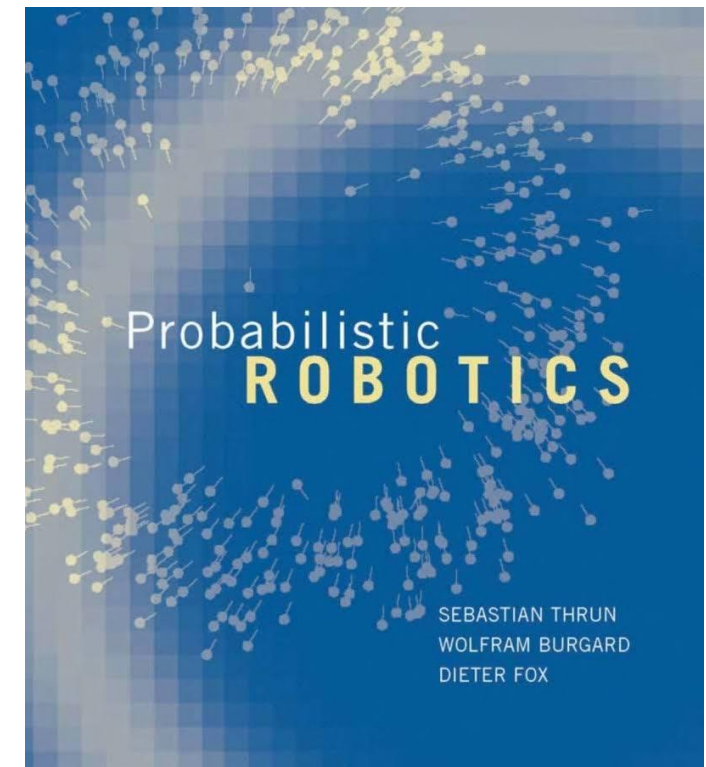
de Claudiu Pozna

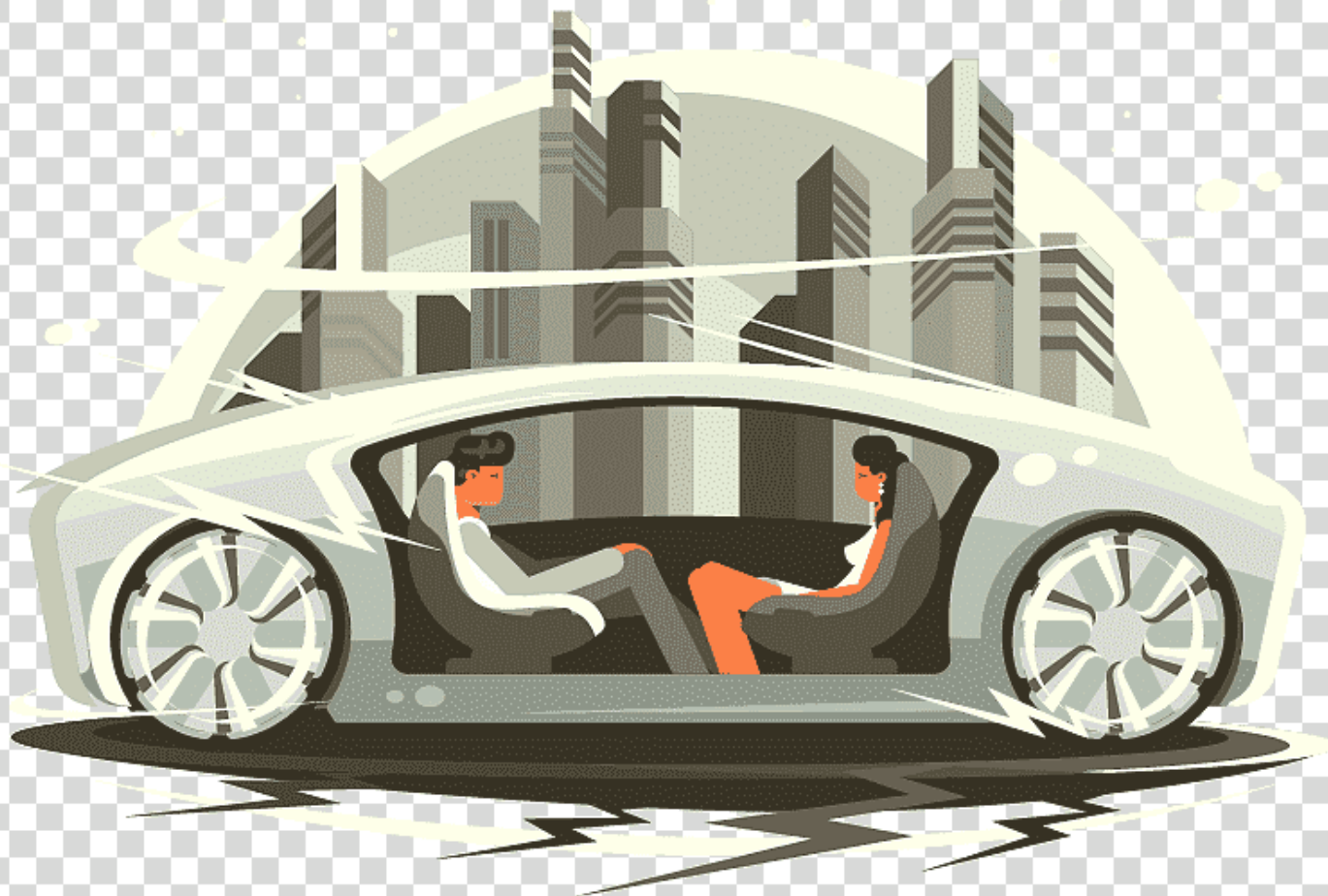
Universitatea Transilvania Brasov

ISBN: 978-606-25-0579-0

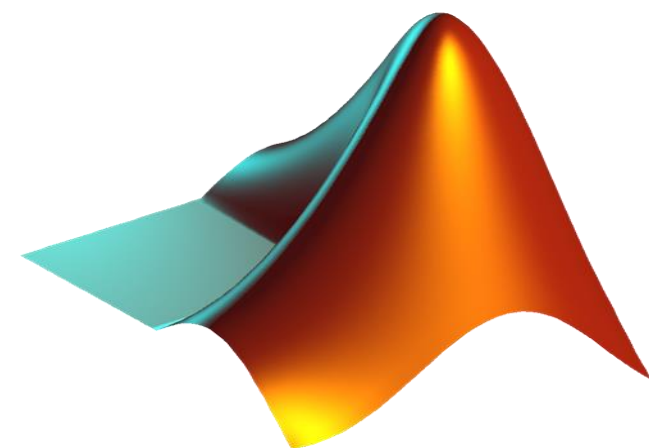
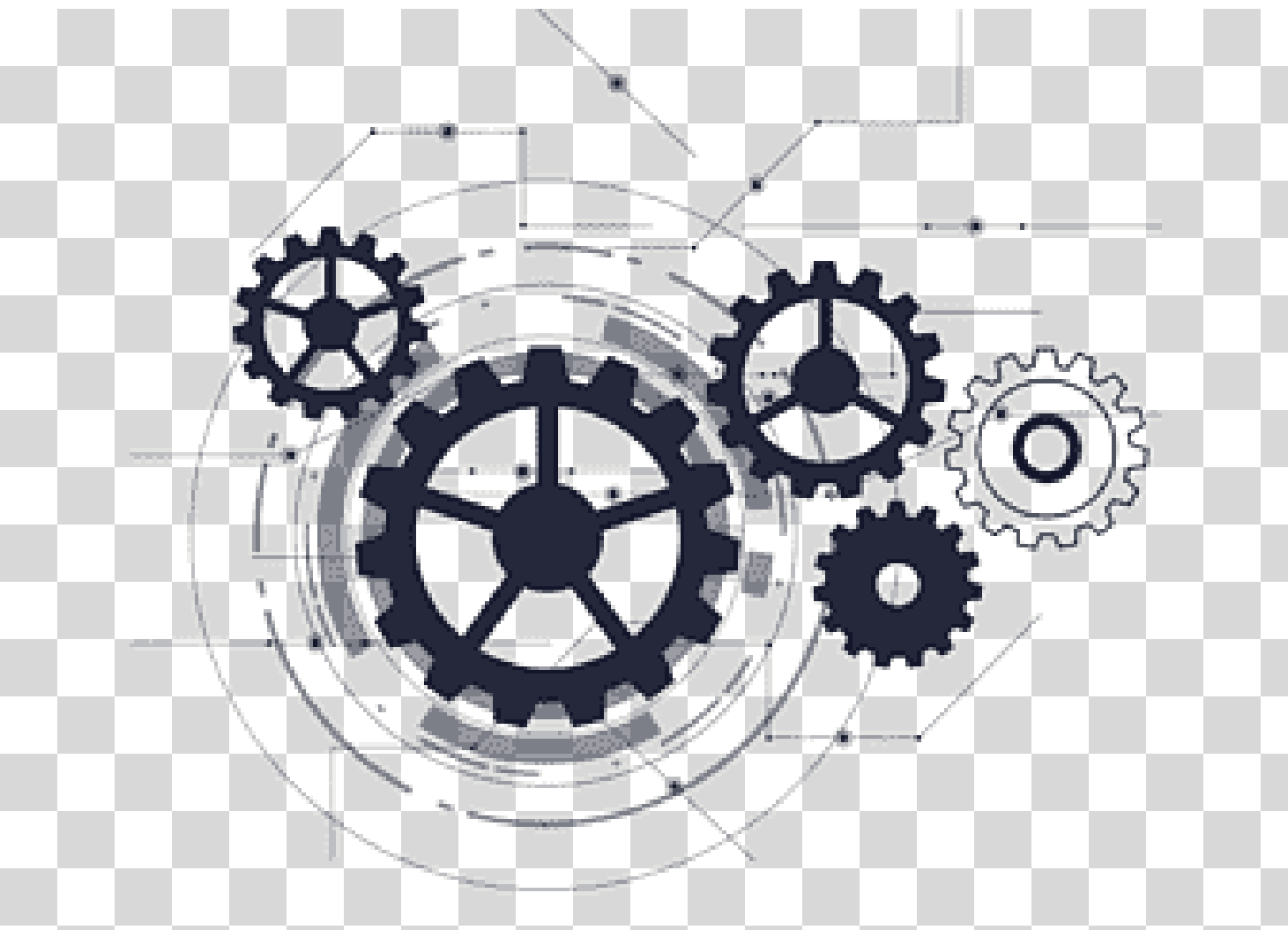
Limba: Romana

Suport: Hartie





Programme utilisezate



Simulări

1. Alegere / Vizualizare

Mediul locomotiei, fisier *.mat

Vizualizare

2. PoliLinie

Punctul i

Xi -100

Yi 95

+

OK

3. Model RM

Viteza [km/h] 36

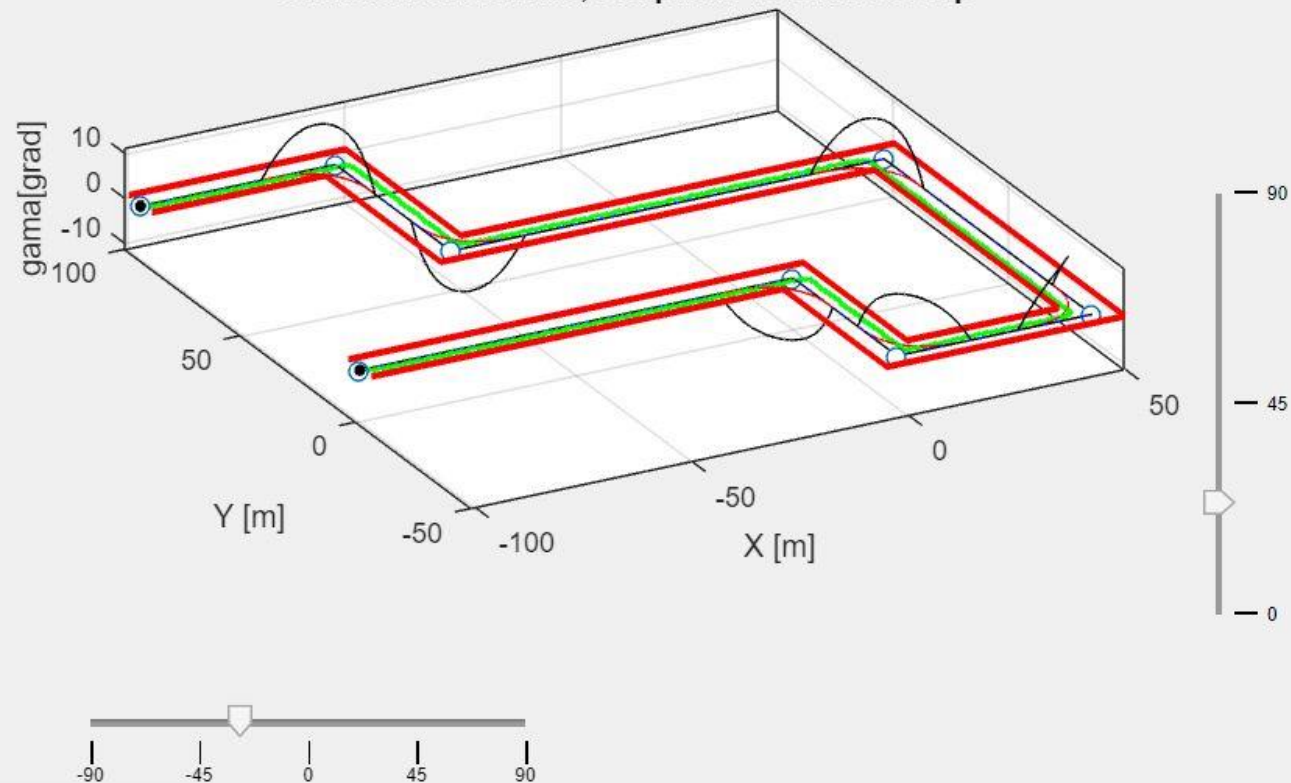
☐ Marimi Cinematice

Simulare / Vizualizare

4. Salvare traiectorie clotoidala

*.mat MediulX.mat

Salvare

Domeniul locomotiei, RM punct de Start si Stop

xmin -100 xmax 100 ymin -100 ymax 100

Date initiale

Traectoria

PID

Kp 26.3

Kv 3.58

Ki 0

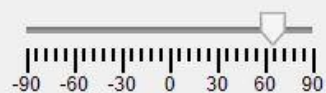
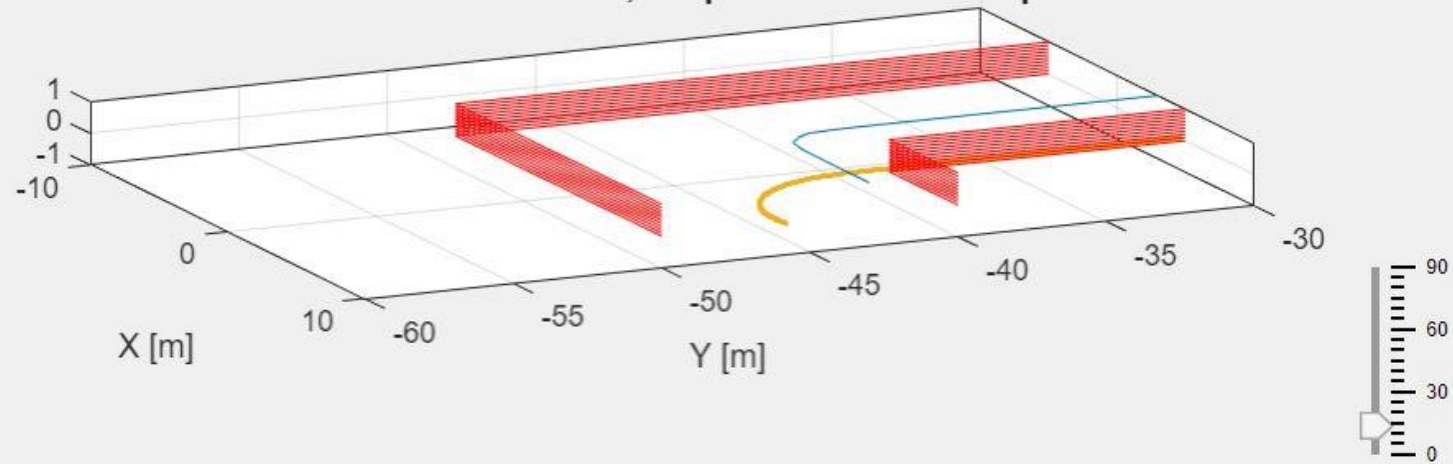
Antrenare

Twiddle

FParticule

Simulare

Domeniul locomotiei, RM punct de Start si Stop



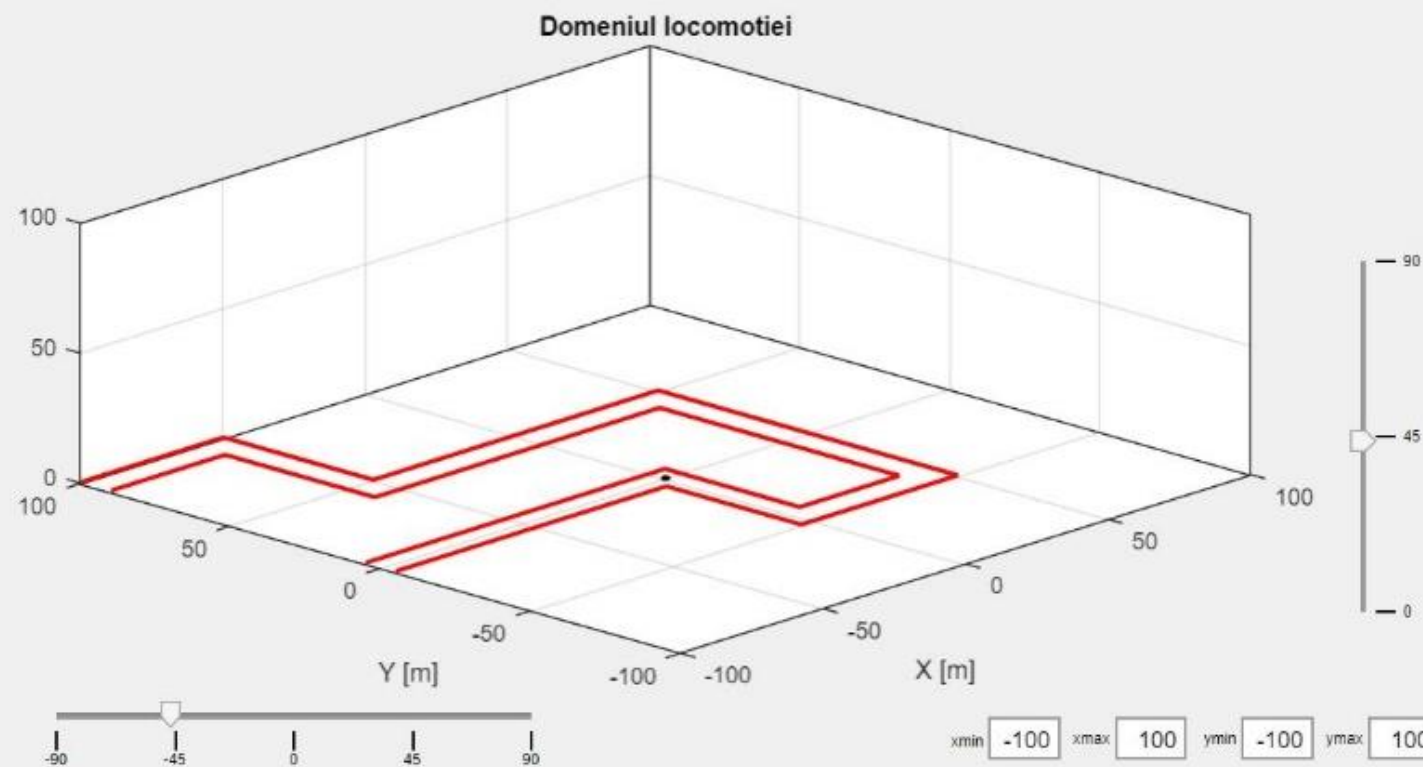
xmin -10

xmax 10

ymin -60

ymax -30

Cla



1. PoliLinie

Punctul i

Xi

Yi

+

OK

2. Discretizarea

dL [m]

3. Finalizare

Nume bD

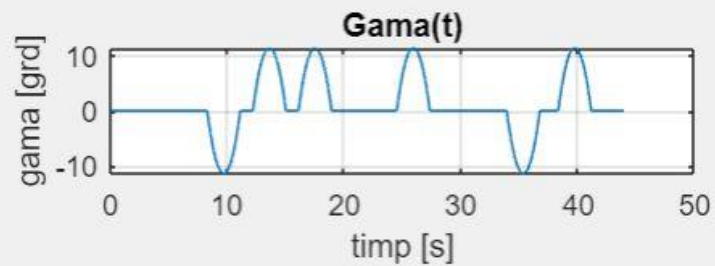
Confirmare

Simularea

1. Alegere / Vizualizare

Mediul locomotiei, fisier *.mat

Vizualizare



Parametrii

Masa [kg]

Jz [kgm²]

K [N/m]

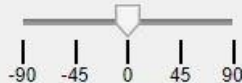
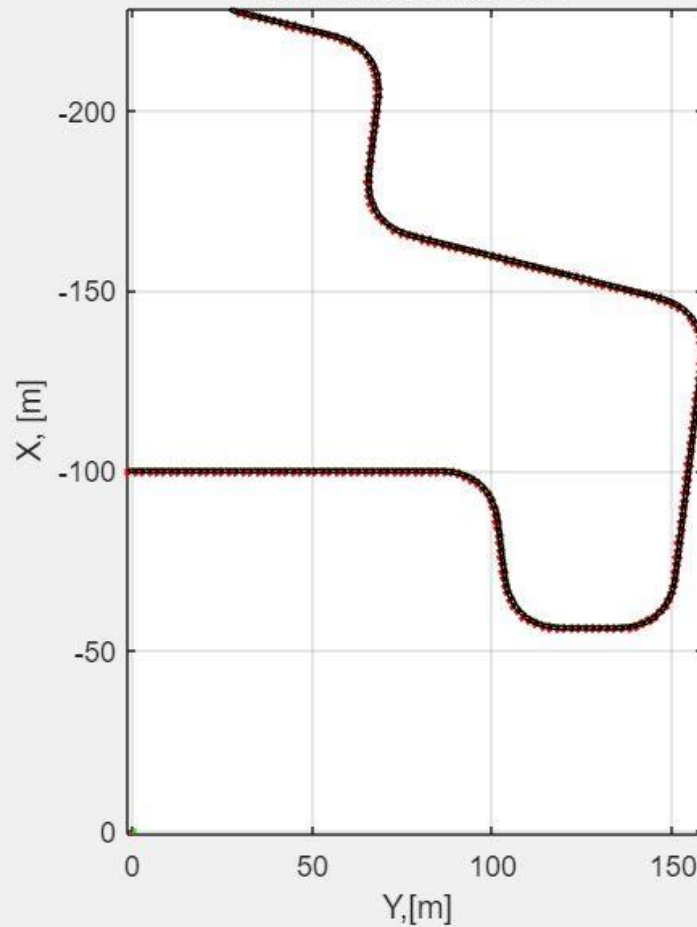
p %

☐ ModCin

☐ Animat

OK

Domeniul locomotiei



xmin xmax ymin ymax



1. Date Intiale

Parametrii Senzorului

Raza [m]

Alfa [grd]

d(Alfa) [grd]

Dom *.mat

2. Simulare 1: Functionare

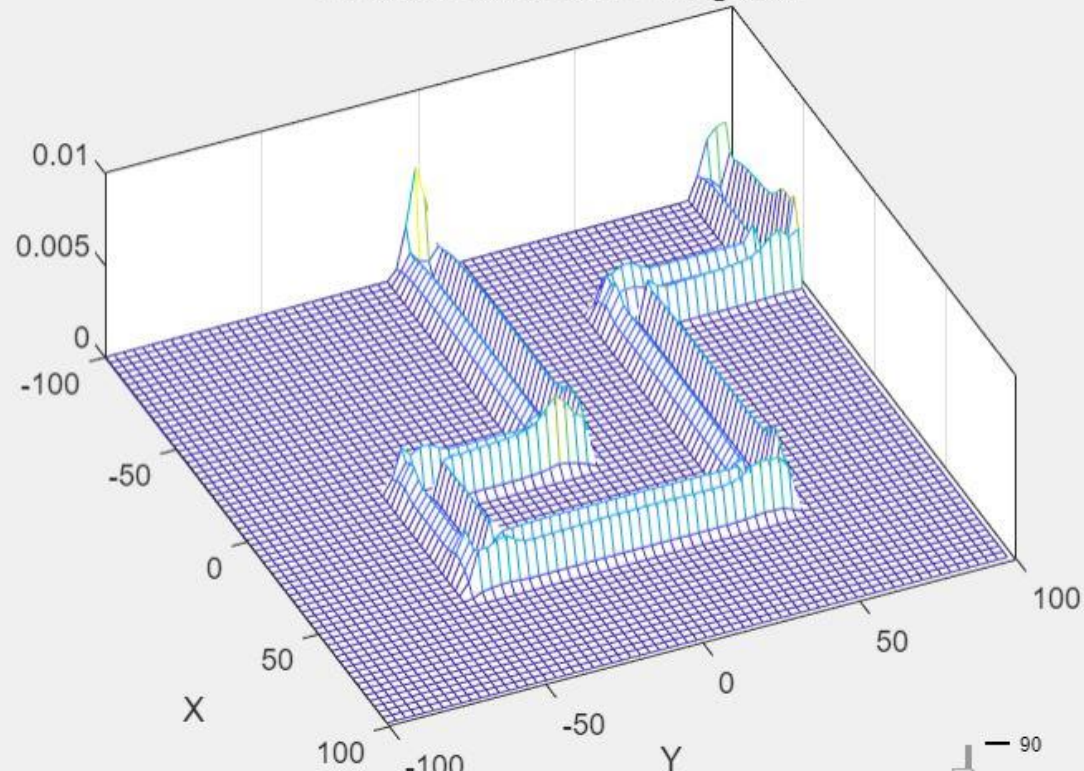
Postura Senzorului

x [m]

y [m]

Teta [grd]

Punctele masurate in SR global



xmin

xmax

ymin

ymax



3. Simulare 2: Masurarea pe Traiectorie



Obiectivele cursului;
Structura cursului;
Bibliografie;
Programe utilizate;
Notare.

Concluzii